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> restart:with(DifferentialGeometry):with(JetCalculus):with(DEtools)
:with(PDEtools):
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```
> DGsetup([t,x],[u,v,w],E,10):
```

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E >
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E > ##### STEP 1
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E > ### component form of the elastic string PDEs
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E > strain:=lambda=sqrt((1+u[2])^2+v[2]^2+w[2]^2)-1;
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$$strain := \lambda = \sqrt{(1+u_2)^2 + v_2^2 + w_2^2} - 1 \quad (1)$$

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```
E > # express k(lambda) as solution of differential equations
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```
E > declare(k(u[2],v[2],w[2]));
```

$k(u_2, v_2, w_2)$ will now be displayed as k (2)

```
E > k_eqns:=[diff(k(u[2],v[2],w[2]),u[2])/(1+u[2])=diff(k(u[2],v[2],w
[2]),v[2])/v[2],diff(k(u[2],v[2],w[2]),u[2])/(1+u[2])=diff(k(u
[2],v[2],w[2]),w[2])/w[2],diff(k(u[2],v[2],w[2]),w[2])/w[2]=diff
(k(u[2],v[2],w[2]),v[2])/v[2]];
```

$$k_eqns := \left[\frac{k_{u_2}}{1+u_2} = \frac{k_{v_2}}{v_2}, \frac{k_{u_2}}{1+u_2} = \frac{k_{w_2}}{w_2}, \frac{k_{w_2}}{w_2} = \frac{k_{v_2}}{v_2} \right] \quad (3)$$

```
E > pdsolve(k_eqns):subs({_F1(2*u[2]+u[2]^2 + v[2]^2 + w[2]^2)=k
(lambda)},%[]):subs(strain,%):k_funct:=%;
```

$$k_funct := k = k\left(\sqrt{(1+u_2)^2 + v_2^2 + w_2^2} - 1\right) \quad (4)$$

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```
E > eqn_u:=u[1,1]-TotalDiff(k(u[2],v[2],w[2])*(1+u[2]),[2]);
```

$$eqn_u := u_{1,1} - \left(k_{u_2} (1+u_2) + k \right) u_{2,2} - k_{v_2} (1+u_2) v_{2,2} - k_{w_2} (1+u_2) w_{2,2} \quad (5)$$

```
E > eqn_v:=v[1,1]-TotalDiff(k(u[2],v[2],w[2])*v[2],[2]);
```

$$eqn_v := v_{1,1} - k_{u_2} v_2 u_{2,2} - \left(k_{v_2} v_2 + k \right) v_{2,2} - k_{w_2} v_2 w_{2,2} \quad (6)$$

```
E > eqn_w:=w[1,1]-TotalDiff(k(u[2],v[2],w[2])*w[2],[2]);
```

$$eqn_w := w_{1,1} - k_{u_2} w_2 u_{2,2} - k_{v_2} w_2 v_{2,2} - \left(k_{w_2} w_2 + k \right) w_{2,2} \quad (7)$$

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```
E > pdes:=[eqn_u,eqn_v,eqn_w]:
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E > solve(pdes,[u[1,1],v[1,1],w[1,1]])[:lead_ders:=%:
```

```
E > ord:=2;
```

$$ord := 2 \quad (8)$$

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E >
E > ### component form of point symmetry
E >
E > dep:=t,x,u[],v[],w[]:
E > P_u:=eta_u(dep)-xi(dep)*u[2]-tau(dep)*u[1];
      
$$P_u := \eta_u(t, x, u[], v[], w[]) - \xi(t, x, u[], v[], w[]) u_2 - \tau(t, x, u[], v[], w[]) u_1 \quad (9)$$


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E > P_v:=eta_v(dep)-xi(dep)*v[2]-tau(dep)*v[1];
      
$$P_v := \eta_v(t, x, u[], v[], w[]) - \xi(t, x, u[], v[], w[]) v_2 - \tau(t, x, u[], v[], w[]) v_1 \quad (10)$$


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E > P_w:=eta_w(dep)-xi(dep)*w[2]-tau(dep)*w[1];
      
$$P_w := \eta_w(t, x, u[], v[], w[]) - \xi(t, x, u[], v[], w[]) w_2 - \tau(t, x, u[], v[], w[]) w_1 \quad (11)$$


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E >
E > symm_vars:=[indets({P_u,P_v,P_w},function)[]]:declare(%):
      eta_u(t,x,u[],v[],w[]) will now be displayed as eta_u
      eta_v(t,x,u[],v[],w[]) will now be displayed as eta_v
      eta_w(t,x,u[],v[],w[]) will now be displayed as eta_w
      tau(t,x,u[],v[],w[]) will now be displayed as tau
      xi(t,x,u[],v[],w[]) will now be displayed as xi
      (12)

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E >
E > symm_generator:=P_u*D_u[]+P_v*D_v[]+P_w*D_w[]:
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E >
E >
E > ##### STEP 2
E >
E > ### determining equations
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E >
E > # apply symmetry generator to each pde
E > symm_pdes:=[seq(Hook(Prolong(symm_generator,ord),
      VerticalExteriorDerivative(pdes[i])),i=1..nops(pdes))):indets(%
      ,name):select(has,%,1):{seq([op(#[i])],i=1..nops(%))}:select(has
      ,%,[1,1],[1,1,1],[1,1,2])));
      {[1,1],[1,1,1],[1,1,2]}
      (13)

```

```

E >
E > # substitute u_tt,v_tt,w_tt, and their t,x derivatives
E > subs(TotalDiff(lead_ders,[1]),symm_pdes):subs(TotalDiff
      (lead_ders,[2]),%):subs(lead_ders,%):detcns:=simplify(%):nops(%)
      ;

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3 (14)

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E > indets(detcns,name):select(has,%,1):{seq([op(#[i])],i=1..nops(%))}

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)} intersect {[1,1],[1,1,1],[1,1,2]};
```

$\emptyset$

(15)

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```
E > # split in jet space (derivatives of u,v,w) excluding (u_x,v_x,
      w_x) in k
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```
E > split_vars:={ (indets({deteqns},name) minus indets({symm_vars,
      strain},name)) []};
```

```
split_vars := {u1, u1,2, u2,2, u1,2,2, u2,2,2, v1, v1,2, v2,2, v1,2,2, v2,2,2, w1, w1,2, w2,2, w1,2,2, w2,2,2}
```

(16)

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```
E > {seq(coeffs(expand(deteqns[i]),split_vars),i=1..nops(deteqns))}
      :detsys:=simplify(%):nops(%);#indets(%%);
```

84

(17)

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E > # determining system of 84 equations
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E > ### conditions on unknowns
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E > # at least one component of the symmetry must be non-zero
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E > symm_conds:=[add(symm_vars[i]^2,i=1..nops(symm_vars))<>0];
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symm_conds := [eta_u^2 + eta_v^2 + eta_w^2 + τ^2 + ξ^2 ≠ 0]
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E > # k(lambda) must be non-constant (otherwise the elastic string
      PDEs are linear)
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E > classify_vars:=[k(u[2],v[2],w[2])];
```

```
classify_vars := [k]
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E > classify_conds:=[diff(k(u[2],v[2],w[2]),u[2])^2 +diff(k(u[2],v
      [2],w[2]),v[2])^2+diff(k(u[2],v[2],w[2]),w[2])^2<>0];
```

```
classify_conds := [ku2 + kv2 + kw2 ≠ 0]
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(20)

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E > ##### STEP 3
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E > ### bring determining system into involutive form
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E > fullsys:=[detsys[],classify_conds[],symm_conds]:nops(%);
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(21)

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E > fulldetsys:=sort([detsys[],k_eqns[],classify_conds[],symm_conds[]
],length):nops(%);
```

89

(22)

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E > split_sys:=rifsimp(fulldetsys,[symm_vars[],classify_vars],
casesplit):
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```
E > split_sys[casecount];
```

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(23)

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E > seq(print(i,remove(has,split_sys[i][Solved],symm_vars)),i=1..
split_sys[casecount]);
```

$$\begin{aligned}
& 1, \left[k_{u_2} = \frac{k_{w_2} (1 + u_2)}{w_2}, k_{v_2} = \frac{k_{w_2} v_2}{w_2} \right] \\
& 2, \left[k_{w_2, w_2, w_2} = \frac{2 k k_{w_2, w_2}^2 w_2^2 - k_{w_2}^2 k_{w_2, w_2} w_2^2 - k k_{w_2} k_{w_2, w_2} w_2 + k_{w_2}^3 w_2 - k k_{w_2}^2}{k k_{w_2} w_2^2}, k_{u_2} = \frac{k_{w_2} (1 + u_2)}{w_2}, \right. \\
& \quad \left. k_{v_2} = \frac{k_{w_2} v_2}{w_2} \right] \\
& 3, \left[k_{w_2, w_2} = \frac{1}{k u_2^2 w_2 + k v_2^2 w_2 + k w_2^3 + 2 w_2 k u_2 + w_2 k} \left(k_{w_2}^2 w_2^3 + k_{w_2}^2 w_2 + k k_{w_2} + k_{w_2}^2 v_2^2 w_2 + k k_{w_2} \right. \right. \\
& \quad \left. \left. v_2^2 + k_{w_2}^2 u_2^2 w_2 + k k_{w_2} u_2^2 - k k_{w_2} w_2^2 + 2 k_{w_2}^2 u_2 w_2 + 2 k k_{w_2} u_2 \right), k_{u_2} = \frac{k_{w_2} (1 + u_2)}{w_2}, k_{v_2} = \frac{k_{w_2} v_2}{w_2} \right] \\
& \quad 4, \left[k_{w_2, w_2} = \frac{k_{w_2} (3 k_{w_2} w_2 + 2 k)}{2 k w_2}, k_{u_2} = \frac{k_{w_2} (1 + u_2)}{w_2}, k_{v_2} = \frac{k_{w_2} v_2}{w_2} \right] \\
& 5, \left[k_{u_2} = -\frac{4 (1 + u_2) k}{u_2^2 + v_2^2 + w_2^2 + 2 u_2 + 1}, k_{v_2} = -\frac{4 v_2 k}{u_2^2 + v_2^2 + w_2^2 + 2 u_2 + 1}, k_{w_2} \right. \\
& \quad \left. = \frac{4 w_2 k}{-u_2^2 - v_2^2 - w_2^2 - 2 u_2 - 1} \right] \\
& \quad 6, \left[k_{w_2, w_2} = \frac{k_{w_2} (3 k_{w_2} w_2 + k)}{k w_2}, k_{u_2} = \frac{k_{w_2} (1 + u_2)}{w_2}, k_{v_2} = \frac{k_{w_2} v_2}{w_2} \right] \\
& 7, \left[k_{u_2} = -\frac{(1 + u_2) k}{u_2^2 + v_2^2 + w_2^2 + 2 u_2 + 1}, k_{v_2} = -\frac{v_2 k}{u_2^2 + v_2^2 + w_2^2 + 2 u_2 + 1}, k_{w_2} = \right. \\
& \quad \left. -\frac{w_2 k}{u_2^2 + v_2^2 + w_2^2 + 2 u_2 + 1} \right]
\end{aligned}$$

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E > ### remove redundant cases
E > dsubs(split_sys[2][Solved],split_sys[1][Solved]):{%[]} minus
      {k_eqns[]}:{seq(lhs(%[i])-rhs(%[i]),i=1..nops(%))}:indets(%);
       $\emptyset$ 

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E > # 2 is special case of 1
E >
E > dsubs(split_sys[3][Solved],split_sys[1][Solved]):{%[]} minus
      {k_eqns[]}:{seq(lhs(%[i])-rhs(%[i]),i=1..nops(%))}:indets(%);
       $\{t, x, u[ ], u_2, v[ ], v_2, w[ ], w_2, k_w, \tau_p, \xi_x, k, \tau, \xi\}$ 

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E > # 3 is new
E >
E > dsubs(split_sys[4][Solved],split_sys[1][Solved]):{%[]} minus
      {k_eqns[]}:{seq(lhs(%[i])-rhs(%[i]),i=1..nops(%))}:indets(%);
       $\emptyset$ 

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E >
E >
E > # 4 is special case of 1
E >
E > dsubs(split_sys[5][Solved],split_sys[1][Solved]):{%[]} minus
      {k_eqns[]}:{seq(lhs(%[i])-rhs(%[i]),i=1..nops(%))}:indets(%);
       $\left\{0, \tau_t - \xi_x \frac{\tau_t}{2} - \frac{\xi_x}{2}\right\}$ 
       $\{t, x, u[ ], v[ ], w[ ], \tau_p, \xi_x, \tau, \xi\}$ 

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E > dsubs(split_sys[5][Solved],split_sys[3][Solved]):{%[]} minus
      {k_eqns[]}:{seq(lhs(%[i])-rhs(%[i]),i=1..nops(%))}:indets(%);
       $\{t, x, u[ ], v[ ], w[ ], \tau_t, \tau_p, \tau\}$ 

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E > # 5 is new
E >
E > dsubs(split_sys[6][Solved],split_sys[1][Solved]):{%[]} minus
      {k_eqns[]}:{seq(lhs(%[i])-rhs(%[i]),i=1..nops(%))}:indets(%);
      {0}
       $\emptyset$ 

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(30)

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E > # 6 is special case of 1
E >
E > dsubs(split_sys[7][Solved],split_sys[1][Solved]):{%[]} minus
      {k_eqns[]}:{seq(lhs(%[i])-rhs(%[i]),i=1..nops(%))}:indets(%);
       $\{0, \tau_t - \xi_x 2 \tau_t - 2 \xi_x\}$ 
       $\{t, x, u[ ], v[ ], w[ ], \tau_p, \xi_x, \tau, \xi\}$ 

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E > dsubs(split_sys[7][Solved],split_sys[3][Solved]):{%[]} minus
      {k_eqns[]}:{seq(lhs(%[i])-rhs(%[i]),i=1..nops(%))}:indets(%);

```

$\{0\}$

$\emptyset$

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E > # 7 is special case of 3

E >

E >

E >

E > ### List of solution CASES: 1,3,5

E >

E >

E > ##### STEP 4

E >

E > ### Do for each case:

E > # first look at the case output and see if it contains a
constraint

E > # next look at case equations and separate into algebraic
conditions, ODEs, PDEs

E > # solve for k first

E > # solve simplest eqns (...=0) next

E > # solve remaining eqns

E >

E >

E >

E > ### CASE 1

E >

E > split_sys[1][Constraint];

$(split_sys_1)_{Constraint}$

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E > split_sys[1][Solved]:sys1:=simplify(%);nops(%);

$$\begin{aligned} sys1 := & \left[\begin{aligned} & eta_u_{t,t} = 0, eta_u_x = 0, eta_u_{u[]} = \xi_x, eta_u_{v[]} = -eta_v_{u[]}, eta_u_{w[]} = -eta_w_{u[]}, eta_v_{t,t} \\ & = 0, eta_v_{t,u[]} = 0, eta_v_{u[],u[]} = 0, eta_v_x = eta_v_{u[]}, eta_v_{v[]} = \xi_x, eta_v_{w[]} = -eta_w_{v[]}, eta_w_{t,t} \\ & = 0, eta_w_{t,u[]} = 0, eta_w_{t,v[]} = 0, eta_w_{u[],u[]} = 0, eta_w_{u[],v[]} = 0, eta_w_{v[],v[]} = 0, eta_w_x \\ & = eta_w_{u[]}, eta_w_{w[]} = \xi_x, \tau_t = \xi_x, \tau_x = 0, \tau_{u[]} = 0, \tau_{v[]} = 0, \tau_{w[]} = 0, \xi_{x,x} = 0, \xi_t = 0, \xi_{u[]} = 0, \xi_{v[]} = 0, \end{aligned} \right. \end{aligned}$$

$$\xi_{w[]} = 0, k_{u_2} = \frac{k_{w_2} (1 + u_2)}{w_2}, k_{v_2} = \frac{k_{w_2} v_2}{w_2} \left[\right.$$

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E >

E > remove(has,sys1,symm_vars);dsubs(k_func,%) : {seq(simplify(lhs(%
[i]))-rhs(%[i])),i=1..nops(%))};

$$\left[k_{u_2} = \frac{k_{w_2} (1 + u_2)}{w_2}, k_{v_2} = \frac{k_{w_2} v_2}{w_2} \right]$$

{0}

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E > k_soln1:=k_func;

$$k_soln1 := k = k(\sqrt{(1 + u_2)^2 + v_2^2 + w_2^2} - 1)$$

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E > remove(has,sys1,k(u[2],v[2],w[2])):sysleqns1:=sort(%,length);nops
(%);

$$\begin{aligned} sysleqns1 := & \left[\xi_t = 0, \tau_x = 0, eta_u_x = 0, \xi_{u[]} = 0, \xi_{v[]} = 0, \xi_{w[]} = 0, \tau_{u[]} = 0, \tau_{v[]} = 0, \tau_{w[]} = 0, \xi_{x,x} = 0, \right. \\ & eta_u_{t,t} = 0, eta_v_{t,t} = 0, eta_w_{t,t} = 0, eta_v_{t,u[]} = 0, eta_w_{t,u[]} = 0, eta_w_{t,v[]} = 0, eta_v_{u[],u[]} = 0, \\ & eta_w_{u[],u[]} = 0, eta_w_{u[],v[]} = 0, eta_w_{v[],v[]} = 0, \tau_t = \xi_x, eta_u_{u[]} = \xi_x, eta_v_{v[]} = \xi_x, eta_w_{w[]} = \\ & = \xi_x, eta_v_x = eta_v_{u[]}, eta_w_x = eta_w_{u[]}, eta_u_{v[]} = -eta_v_{u[]}, eta_u_{w[]} = -eta_w_{u[]}, eta_v_{w[]} = \\ & \left. -eta_w_{v[]} \right] \end{aligned}$$

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E >

E > select(has,sysleqns1,0):pdsolve(%) : syslsoln1:=simplify(%);

$$\begin{aligned} syslsoln1 := & \left\{ eta_u = f_{15}(u[], v[], w[]) t + f_{16}(u[], v[], w[]), eta_v = f_{12}(x, v[], w[]) t + f_{13}(x, \right. \\ & v[], w[]) u[] + f_{14}(x, v[], w[]), eta_w = f_3(x, w[]) t + f_6(x, w[]) u[] + f_7(x, w[]) v[] + f_8(x, \\ & \left. w[]), \tau = f_9(t), \xi = c_1 x + c_2 \right\} \end{aligned}$$

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E > map(rhs,syslsoln1):indets(%,name) minus indets(symm_vars,name)
:syslCs1:=%;indets(%%,name) minus indets(indets(%%,function),
name) minus %;

$$syslCs1 := \{c_1, c_2\}$$

$\emptyset$

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E > dsubs(syslsoln1,sysleqns1) : {seq(coeffs(expand(lhs(%[i]))-rhs(%[i]))
,[]),i=1..nops(%))}:rifsimp(%,casesplit):simplify(%[Solved])
:sysleqns2:=%;nops(%);

$$sysleqns2 := \left[f_{12_x} = 0, f_{13_x} = 0, f_{14_x} = f_{13}(x, v[], w[]), f_{3_x} = 0, f_{6_x} = 0, f_{7_x} = 0, f_{8_x} = f_6(x, w[]), f_{15_{u[]}} = 0, \right.$$

$$\begin{aligned}
f_{16}_{u[]} &= c_P f_{12}_{v[]} = 0, f_{13}_{v[]} = 0, f_{14}_{v[]} = c_P f_{15}_{v[]} = 0, f_{16}_{v[]} = -f_{13}(x, v[], w[]), f_{12}_{w[]} = 0, f_{13}_{w[]} \\
&= 0, f_{14}_{w[]} = -f_7(x, w[]), f_{15}_{w[]} = 0, f_{16}_{w[]} = -f_6(x, w[]), f_3_{w[]} = 0, f_6_{w[]} = 0, f_7_{w[]} = 0, f_8_{w[]} = c_P, \\
f_9_t &= c_I
\end{aligned}$$

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E > select(has,sysleqns2,0);pdsolve(%);subs({c__1=c__11,c__2=c__12},
%):syslsoln2:=simplify(%);
[f12x=0,f13x=0,f3x=0,f6x=0,f7x=0,f15u[ ]=0,f12v[ ]=0,f13v[ ]=0,f15v[ ]=0,f12w[ ]=0,f13w[ ]=0,
f15w[ ]=0,f3w[ ]=0,f6w[ ]=0,f7w[ ]=0]
{f12(x,v[ ],w[ ]) = c1,f13(x,v[ ],w[ ]) = c2,f15(u[ ],v[ ],w[ ]) = c3,f3(x,w[ ]) = c4,f6(x,w[ ]) = c5,f7(x,
w[ ]) = c6}
syslsoln2 := {f12(x,v[ ],w[ ]) = c11,f13(x,v[ ],w[ ]) = c12,f15(u[ ],v[ ],w[ ]) = c3,f3(x,w[ ]) = c4,f6(x, (41)
w[ ]) = c5,f7(x,w[ ]) = c6}

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E > map(rhs,syslsoln2):indets(%,name) minus indets(symm_vars,name)
:syslCs2:=% union syslCs1;indets(%%,name) minus indets(indets
(%%,function),name) minus %;
syslCs2 := {c1,c11,c12,c2,c3,c4,c5,c6}
Ø
(42)

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E > dsubs(syslsoln2,sysleqns2):{seq(coeffs(expand(lhs(%[i])-rhs(%[i])
),[ ]),i=1..nops(%))}:rifsimp(%,casesplit):simplify(%[Solved])
:sysleqns3:=%;nops(%);
sysleqns3 := [f14x=c12,f8x=c5,f16u[ ]=c1,f14v[ ]=c1,f16v[ ]=-c12,f14w[ ]=-c6,f16w[ ]=-c5,f8w[ ]
=c1,f9t=c1]
9
(43)

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E > sysleqns3:pdsolve(%):syslsoln3:=simplify(%);
syslsoln3 := {f14(x,v[ ],w[ ]) = c1 v[ ] + c12 x - c6 w[ ] + c13, f16(u[ ],v[ ],w[ ]) = c1 u[ ] - c12 v[ ] (44)
- c5 w[ ] + c14, f8(x,w[ ]) = c1 w[ ] + c5 x + c15, f9(t) = c1 t + c16}

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E > map(rhs,syslsoln3):indets(%,name) minus indets(symm_vars,name)
:syslCs3:=% union syslCs2;#indets(%%,name) minus indets(indets
(%%,function),name) minus %;
syslCs3 := {c1,c11,c12,c13,c14,c15,c16,c2,c3,c4,c5,c6}
(45)

```

```

E > # all eqns solved
E > syslsoln1:dsubs(syslsoln2,%):dsubs(syslsoln3,%):soln1:=simplify(
[%[]]);
soln1 := [eta_u=c1 u[ ] - c12 v[ ] + c3 t - c5 w[ ] + c14, eta_v=(x+u[ ]) c12 + c11 t - c6 w[ ] (46)
+ c1 v[ ] + c13, eta_w=(x+u[ ]) c5 + c4 t + c1 w[ ] + c6 v[ ] + c15, tau=c1 t + c16, xi=c1 x + c2]

```



```
E >
E > dsubs(soln1,detsys):dsubs(k_soln1,%);
                                {0}
(47)
```

```
E > # verified
```

```
E >
```

```
E > map(rhs,soln1):indets(% ,name) minus indets(symm_vars,name)
:soln1Cs:=%;nops(%);
                                soln1Cs := {c1, c11, c12, c13, c14, c15, c16, c2, c3, c4, c5, c6}
                                12
(48)
```

```
E >
```

```
E > k_soln1:subs(solve(strain,{w[2]^2}),rhs(%)):simplify(%) assuming
(lambda>0);
```

Warning, solving for expressions other than names or functions is not recommended.

$k(\lambda)$ (49)

```
E >
```

```
E >
```

```
E >
```

```
E > ### CASE 3
```

```
E > split_sys[3][Pivots]:sys3conds:=%;
```

```
sys3conds := [kw2 ≠ 0, eta-u2 + eta-v2 + eta-w2 + τ2 + ξ2 ≠ 0, kw2 u22 + kw2 v22 + w22 kw2 + w2 k
+ 2 kw2 u2 + kw2 ≠ 0, kw2 u22 + kw2 v22 + w22 kw2 + 4 w2 k + 2 kw2 u2 + kw2 ≠ 0]
```

```
E > split_sys[3][Constraint];
```

$(split_sys_3)_{Constraint}$ (51)

```
E > split_sys[3][Solved]:sys3:=simplify(%) ;nops(%);
```

```
sys3 := [eta-ut,t = 0, eta-ux =  $\frac{2(-\tau_t + \xi_x) k w_2}{k_{w_2} (u_2^2 + v_2^2 + w_2^2 + 2 u_2 + 1)}$ , eta-uu [ ]
=  $\frac{(k_{w_2} (u_2^2 + v_2^2 + w_2^2 + 2 u_2 + 1) + 2 w_2 k) \xi_x - 2 \tau_t k w_2}{k_{w_2} (u_2^2 + v_2^2 + w_2^2 + 2 u_2 + 1)}$ , eta-uv [ ] = -eta-vu [ ], eta-uw [ ] =
```

$-eta_{-w_{u[]}}, eta_{-v_{t,t}} = 0, eta_{-v_{t,u[]}} = 0, eta_{-v_{u[],u[]}} = 0, eta_{-v_x} = eta_{-v_{u[]}}, eta_{-v_{v[]}}$

$$= \frac{\left(k_{w_2} \left(u_2^2 + v_2^2 + w_2^2 + 2u_2 + 1 \right) + 2w_2 k \right) \xi_x - 2\tau_t k w_2}{k_{w_2} \left(u_2^2 + v_2^2 + w_2^2 + 2u_2 + 1 \right)}, \eta_{w_2} = -\eta_{w_2}, \eta_{w_2} = 0,$$

$$\eta_{w_2} = 0, \eta_{w_2} = 0, \eta_{w_2} = 0, \eta_{w_2} = 0, \eta_{w_2} = 0, \eta_{w_2} = \eta_{w_2},$$

$$\eta_{w_2} = \frac{\left(k_{w_2} \left(u_2^2 + v_2^2 + w_2^2 + 2u_2 + 1 \right) + 2w_2 k \right) \xi_x - 2\tau_t k w_2}{k_{w_2} \left(u_2^2 + v_2^2 + w_2^2 + 2u_2 + 1 \right)}, \tau_{w_2} = 0, \tau_x = 0, \tau_{w_2} = 0, \tau_{w_2}$$

$$= 0, \tau_{w_2} = 0, \xi_{w_2} = 0, \xi_t = 0, \xi_{w_2} = 0, \xi_{w_2} = 0, \xi_{w_2} = 0, k_{w_2, w_2}$$

$$= \frac{\left(w_2 \left(u_2^2 + v_2^2 + w_2^2 + 2u_2 + 1 \right) k_{w_2} + k \left(u_2^2 + v_2^2 - w_2^2 + 2u_2 + 1 \right) \right) k_{w_2}}{k w_2 \left(u_2^2 + v_2^2 + w_2^2 + 2u_2 + 1 \right)}, k_{w_2} = \frac{k_{w_2} (1 + u_2)}{w_2},$$

$$k_{w_2} = \frac{k_{w_2} v_2}{w_2}$$

32

(52)

E >

```
E > remove(has,sys3,symm_vars):dsubs(k_funct,%):{seq(simplify(lhs(%
[i])-rhs(%[i])),i=1..nops(%))}:subs(solve(strain,{w[2]^2}),%)
:simplify(%):assuming(lambda>0):convert(%):{seq(coeffs
(expand(%[i]),[u[2],v[2],w[2]]),i=1..nops(%))} minus {0}
:sys3_keqns:=simplify(%):nops(%);
```

Warning, solving for expressions other than names or functions is not recommended.

$$\text{sys3_keqns} := \left\{ \frac{-2k(\lambda)(\lambda+1)k_{\lambda,\lambda} + 2k_{\lambda}((\lambda+1)k_{\lambda} - k(\lambda))}{(\lambda+1)^3 k(\lambda)}, \right. \\ \frac{-k(\lambda)(\lambda+1)k_{\lambda,\lambda} + k_{\lambda}((\lambda+1)k_{\lambda} - k(\lambda))}{(\lambda+1)^3 k(\lambda)}, \\ \left. \frac{(k(\lambda)(\lambda+1)k_{\lambda,\lambda} - k_{\lambda}((\lambda+1)k_{\lambda} - k(\lambda)))\lambda(\lambda+2)}{(\lambda+1)^3 k(\lambda)} \right\}$$

3

(53)

```
E > rifsimp({sys3_keqns[],diff(k(lambda),lambda)<>0},casesplit)
:simplify(%[Solved]);dsolve(%)[]:subs({_C2=k0,_C1=p},%):sys3_k:=%
[];
```

$$\left[k_{\lambda,\lambda} = \frac{k_{\lambda} \left((\lambda + 1) k_{\lambda} - k(\lambda) \right)}{k(\lambda) (\lambda + 1)} \right]$$

$$\text{sys3_k} := k(\lambda) = (\lambda + 1)^p k_0 \quad (54)$$

```
E > k_soln3:=k(u[2],v[2],w[2])=subs(strain,rhs(sys3_k));
```

$$k_soln3 := k = \left(\sqrt{(1 + u_2)^2 + v_2^2 + w_2^2} \right)^p k_0 \quad (55)$$

```
E > dsubs(k_soln3,sys3conds):simplify(%) :select(has,%,p):solve(%,{p})
:sys3_p:=%;
```

$$\text{sys3_p} := \{p \neq -4, p \neq -1, p \neq 0\} \quad (56)$$

```
E > select(has,sys3,symm_vars):dsubs(k_soln3,%) :simplify(%)
:sys3eqns1:=sort(%,length);nops(%) ;
```

$$\text{sys3eqns1} := \left[\begin{aligned} &\xi_t = 0, \tau_x = 0, \xi_{u[]} = 0, \xi_{v[]} = 0, \xi_{w[]} = 0, \tau_{u[]} = 0, \tau_{v[]} = 0, \tau_{w[]} = 0, \xi_{x,x} = 0, \tau_{t,t} = 0, \\ &\eta_{u,t} = 0, \eta_{v,t} = 0, \eta_{w,t} = 0, \eta_{v,t,u[]} = 0, \eta_{w,t,u[]} = 0, \eta_{w,t,v[]} = 0, \eta_{v,u[],u[]} = 0, \\ &\eta_{w,u[],u[]} = 0, \eta_{w,u[],v[]} = 0, \eta_{w,v[],v[]} = 0, \eta_{v,x} = \eta_{v,u[]}, \eta_{w,x} = \eta_{w,u[]}, \eta_{u,v[]} = \\ &-\eta_{v,u[]} , \eta_{u,w[]} = -\eta_{w,u[]} , \eta_{v,w[]} = -\eta_{w,v[]} , \eta_{u,x} = \frac{-2\tau_t + 2\xi_x}{p}, \eta_{u,u[]} = \\ &= \frac{(p+2)\xi_x - 2\tau_t}{p}, \eta_{v,v[]} = \frac{(p+2)\xi_x - 2\tau_t}{p}, \eta_{w,w[]} = \frac{(p+2)\xi_x - 2\tau_t}{p} \end{aligned} \right]$$

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```
E > select(has,sys3eqns1,0):pdsolve({%[],sys3_p[]}):sys3soln1:=
simplify(%) ;
```

$$\text{sys3soln1} := \left\{ \begin{aligned} &\eta_{u,x} = f_{14}(x, u[], v[], w[]) t + f_{15}(x, u[], v[], w[]) t \\ &+ f_{12}(x, v[], w[]) u[] + f_{13}(x, v[], w[]) , \eta_{v,x} = f_{11}(x, v[], w[]) t \\ &+ f_{12}(x, v[], w[]) u[] + f_{13}(x, v[], w[]) , \eta_{w,x} = f_3(x, w[]) t + f_6(x, w[]) u[] + f_7(x, w[]) v[] \\ &+ f_8(x, w[]) , \tau = c_1 t + c_2, \xi = c_3 x + c_4 \end{aligned} \right\} \quad (58)$$

```
E > map(rhs,sys3soln1):indets(%,name) minus indets(symm_vars,name)
:sys3Cs1:=%;indets(%%%,name) minus indets(indets(%%%,function),
name) minus %;
```

$$\text{sys3Cs1} := \{c_1, c_2, c_3, c_4\}$$

$$\{t\} \quad (59)$$

```
E > dsubs(sys3soln1,sys3eqns1):{seq(coeffs(expand(lhs(%[i])-rhs(%[i]))
),[t]),i=1..nops(%)}) :rifsimp({%[],sys3_p[]},casesplit):simplify
(%[Solved]):sys3eqns2:=%;nops(%) ;
```

$$\begin{aligned}
\text{sys3eqns2} := & \left[f_{11x} = 0, f_{12x} = 0, f_{13x} = f_{12}(x, v[], w[]), f_{14x} = 0, f_{15x} = \frac{-2c_l + 2c_3}{p}, f_{3x} = 0, f_{6x} = 0, f_{7x} \right. \\
& = 0, f_{8x} = f_6(x, w[]), f_{14u[]} = 0, f_{15u[]} = \frac{(p+2)c_3 - 2c_l}{p}, f_{11v[]} = 0, f_{12v[]} = 0, f_{13v[]} \\
& = \frac{(p+2)c_3 - 2c_l}{p}, f_{14v[]} = 0, f_{15v[]} = -f_{12}(x, v[], w[]), f_{11w[]} = 0, f_{12w[]} = 0, f_{13w[]} = -f_7(x, \\
& w[]), f_{14w[]} = 0, f_{15w[]} = -f_6(x, w[]), f_{3w[]} = 0, f_{6w[]} = 0, f_{7w[]} = 0, f_{8w[]} = \frac{(p+2)c_3 - 2c_l}{p} \left. \right]
\end{aligned}$$

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```

E > select(has, sys3eqns2, 0); pdsolve(%); subs({c__1=c__11, c__2=c__12,
c__3=c__13, c__4=c__14}, %); sys3soln2:=simplify(%);

```

$$\begin{aligned}
& [f_{11x} = 0, f_{12x} = 0, f_{14x} = 0, f_{3x} = 0, f_{6x} = 0, f_{7x} = 0, f_{14u[]} = 0, f_{11v[]} = 0, f_{12v[]} = 0, f_{14v[]} = 0, f_{11w[]} = 0, f_{12w[]} \\
& = 0, f_{14w[]} = 0, f_{3w[]} = 0, f_{6w[]} = 0, f_{7w[]} = 0] \\
& \{f_{11}(x, v[], w[]) = c_2, f_{12}(x, v[], w[]) = c_3, f_{14}(x, u[], v[], w[]) = c_l, f_3(x, w[]) = c_4, f_6(x, w[]) = c_5, \\
& f_7(x, w[]) = c_6\} \\
& \{f_{11}(x, v[], w[]) = c_{12}, f_{12}(x, v[], w[]) = c_{13}, f_{14}(x, u[], v[], w[]) = c_{1l}, f_3(x, w[]) = c_{14}, f_6(x, w[]) \\
& = c_5, f_7(x, w[]) = c_6\}
\end{aligned}$$

```

sys3soln2 := {f_{11}(x, v[], w[]) = c_{12}, f_{12}(x, v[], w[]) = c_{13}, f_{14}(x, u[], v[], w[]) = c_{1l}, f_3(x, w[]) = c_{14},
f_6(x, w[]) = c_5, f_7(x, w[]) = c_6}

```

(61)

```

E > map(rhs, sys3soln2): indets(%, name) minus indets(symm_vars, name)
: sys3Cs2 := % union sys3Cs1; indets(%%, name) minus indets(indets
(%%, function), name) minus %;

```

$$\text{sys3Cs2} := \{c_l, c_{1l}, c_{12}, c_{13}, c_{14}, c_2, c_3, c_4, c_5, c_6\}$$

$\emptyset$

(62)

```

E > dsubs(sys3soln2, sys3eqns2): {seq(expand(lhs(%[i]) - rhs(%[i])), i=1..
nops(%))}: rifsimp({%[], sys3_p[]}, casesplit): simplify(%[Solved])
: sys3eqns3 := %; nops(%); #dsubs(sys3soln2, sys3eqns2): {seq(coeffs
(expand(lhs(%[i]) - rhs(%[i])), [t]), i=1..nops(%))}: rifsimp({%[],
sys3_p[]}, casesplit): simplify(%[Solved]): sys3eqns3 := %; nops(%);

```

$$\begin{aligned}
\text{sys3eqns3} := & \left[f_{13x} = c_{13}, f_{15x} = \frac{-2c_l + 2c_3}{p}, f_{8x} = c_5, f_{15u[]} = \frac{(p+2)c_3 - 2c_l}{p}, f_{13v[]} \right. \\
& = \frac{(p+2)c_3 - 2c_l}{p}, f_{15v[]} = -c_{13}, f_{13w[]} = -c_6, f_{15w[]} = -c_5, f_{8w[]} = \frac{(p+2)c_3 - 2c_l}{p} \left. \right]
\end{aligned}$$

9

(63)

```

E > sys3eqns3: pdsolve({%[], sys3_p[]}): subs({c__14=c__24}, %)
: sys3soln3:=simplify(%);

```

$$\text{sys3soln3} := \left\{ f_{13}(x, v[], w[]) = c_{13}x - c_6 w[] + \frac{v[]((p+2)c_3 - 2c_l)}{p} + c_{15}f_{15}(x, u[], v[], \right. \quad (64)$$

$$\left. w[]) = \frac{(-c_{13}v[] + c_3u[] - c_5w[] + c_{24})p - 2(c_l - c_3)(x + u[]) }{p}, f_8(x, w[]) = c_5x \right. \\ \left. + \frac{((p+2)c_3 - 2c_l)w[]}{p} + c_{16} \right\}$$

```
E > map(rhs,sys3soln3):indets(% ,name) minus indets({k_soln3,
  symm_vars},name):sys3Cs3:=% union sys3Cs2;nops(%);
  sys3Cs3 := {c_l, c_{11}, c_{12}, c_{13}, c_{14}, c_{15}, c_{16}, c_2, c_{24}, c_3, c_4, c_5, c_6}
```

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(65)

```
E > # all eqns solved
```

```
E >
```

```
E > sys3soln1:dsubs(sys3soln2,%):dsubs(sys3soln3,%):soln3:=simplify(
  [%[]]);
```

$$\text{soln3} := \left[\begin{aligned} \eta_u &= \frac{(c_{11}t - c_{13}v[] + c_3u[] - c_5w[] + c_{24})p - 2(c_l - c_3)(x + u[]) }{p}, \eta_v \\ &= \frac{(c_3v[] + (x + u[])c_{13} - c_6w[] + c_{12}t + c_{15})p - 2v[] (c_l - c_3)}{p}, \eta_w \\ &= \frac{(c_3w[] + (x + u[])c_5 + c_6v[] + c_{14}t + c_{16})p - 2w[] (c_l - c_3)}{p}, \tau = c_l t + c_2, \xi = c_3 x \\ &+ c_4 \end{aligned} \right] \quad (66)$$

```
E > dsubs(soln3,detsys):dsubs(k_soln3,%);
```

{0}

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```
E > # verified
```

```
E >
```

```
E > map(rhs,soln3):indets(% ,name) minus indets({symm_vars,k_soln3},
  name):soln3Cs:=%;nops(%);
```

$$\text{soln3Cs} := \{c_l, c_{11}, c_{12}, c_{13}, c_{14}, c_{15}, c_{16}, c_2, c_{24}, c_3, c_4, c_5, c_6\}$$

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(68)

```
E >
```

```
E > k_soln3;sys3_p;
```

$$k = \left(\sqrt{(1 + u_2)^2 + v_2^2 + w_2^2} \right)^p k_0 \\ \{p \neq -4, p \neq -1, p \neq 0\}$$

(69)

```
E >
```

```
E >
```

```
E >
```

E > ### CASE 5

E >

E > split_sys[5][Pivots]:sys5conds:=%;

$$\text{sys5conds} := [k \neq 0, \text{eta_u}^2 + \text{eta_v}^2 + \text{eta_w}^2 + \tau^2 + \xi^2 \neq 0] \quad (70)$$

E > split_sys[5][Constraint];

$$(\text{split_sys5})_{\text{Constraint}} \quad (71)$$

E > split_sys[5][Solved]:sys5:=simplify(%);nops(%);

$$\begin{aligned} \text{sys5} := & \left[\text{eta_u}_{t,t} = 0, \text{eta_u}_x = \frac{\tau_t}{2} - \frac{\xi_x}{2}, \text{eta_u}_{u[]} = \frac{\tau_t}{2} + \frac{\xi_x}{2}, \text{eta_u}_{v[]} = -\text{eta_v}_{u[]}, \text{eta_u}_{w[]} = \right. \\ & -\text{eta_w}_{u[]} , \text{eta_v}_{t,t} = 0, \text{eta_v}_{t,u[]} = 0, \text{eta_v}_{u[],u[]} = 0, \text{eta_v}_x = \text{eta_v}_{u[]} , \text{eta_v}_{v[]} = \frac{\tau_t}{2} + \frac{\xi_x}{2}, \\ & \text{eta_v}_{w[]} = -\text{eta_w}_{v[]} , \text{eta_w}_{t,t} = 0, \text{eta_w}_{t,u[]} = 0, \text{eta_w}_{t,v[]} = 0, \text{eta_w}_{u[],u[]} = 0, \text{eta_w}_{u[],v[]} = 0, \\ & \text{eta_w}_{v[],v[]} = 0, \text{eta_w}_x = \text{eta_w}_{u[]} , \text{eta_w}_{w[]} = \frac{\tau_t}{2} + \frac{\xi_x}{2}, \tau_{t,t,t} = 0, \tau_x = 0, \tau_{u[]} = 0, \tau_{v[]} = 0, \tau_{w[]} \\ & = 0, \xi_{x,x} = 0, \xi_t = 0, \xi_{u[]} = 0, \xi_{v[]} = 0, \xi_{w[]} = 0, k_{u_2} = -\frac{4(1+u_2)k}{u_2^2 + v_2^2 + w_2^2 + 2u_2 + 1}, k_{v_2} = \\ & \left. -\frac{4v_2k}{u_2^2 + v_2^2 + w_2^2 + 2u_2 + 1}, k_{w_2} = -\frac{4w_2k}{u_2^2 + v_2^2 + w_2^2 + 2u_2 + 1} \right] \end{aligned} \quad (72)$$

E > remove(has,sys5,symm_vars):dsubs(k_funct,%):{seq(simplify(lhs(%
[i])-rhs(%[i])),i=1..nops(%))}:subs(solve(strain,{w[2]^2}),%):
simplify(%):assuming(lambda>0):convert(% ,diff):{seq(coeffs
(expand(%[i]),[u[2],v[2],w[2]]),i=1..nops(%))} minus {0}
:sys5_keqns:=simplify(%);nops(%);

Warning, solving for expressions other than names or functions is not recommended.

$$\begin{aligned} & \left\{ \frac{((\lambda+1)k_\lambda + 4k(\lambda))v_2}{(\lambda+1)^2}, \frac{((\lambda+1)k_\lambda + 4k(\lambda))w_2}{(\lambda+1)^2}, \frac{((\lambda+1)k_\lambda + 4k(\lambda))(1+u_2)}{(\lambda+1)^2} \right\} \\ & \text{sys5_keqns} := \left\{ \frac{(\lambda+1)k_\lambda + 4k(\lambda)}{(\lambda+1)^2} \right\} \end{aligned} \quad (73)$$

E > rifsimp({sys5_keqns[],diff(k(lambda),lambda)<>0},casesplit)
:simplify(%[Solved]);dsolve(%):subs({_C1=k0},%):sys5_k:=%;

$$\left[k_{\lambda} = -\frac{4k(\lambda)}{\lambda + 1} \right]$$

$$\text{sys5_k} := k(\lambda) = \frac{k0}{(\lambda + 1)^4} \quad (74)$$

E > k_soln5:=k(u[2],v[2],w[2])=subs(strain,rhs(sys5_k));

$$k_{\text{soln5}} := k = \frac{k0}{((1 + u_2)^2 + v_2^2 + w_2^2)^2} \quad (75)$$

E > dsubs(k_soln5,sys5conds):simplify(%):select(has,%,k0);

$$\left[\frac{k0}{(u_2^2 + v_2^2 + w_2^2 + 2u_2 + 1)^2} \neq 0 \right] \quad (76)$$

E >

**E > select(has,sys5,symm_vars):simplify(%):sys5eqns1:=sort(%,length);
nops(%);**

$$\text{sys5eqns1} := \left[\xi_t = 0, \tau_x = 0, \xi_{u[]} = 0, \xi_{v[]} = 0, \xi_{w[]} = 0, \tau_{u[]} = 0, \tau_{v[]} = 0, \tau_{w[]} = 0, \xi_{x,x} = 0, \eta_{u,t,t} = 0, \right.$$

$$\eta_{v,t,t} = 0, \eta_{w,t,t} = 0, \eta_{v,t,u[]} = 0, \eta_{w,t,u[]} = 0, \eta_{w,t,v[]} = 0, \eta_{v,u[],u[]} = 0, \eta_{w,u[],u[]}$$

$$= 0, \eta_{w,u[],v[]} = 0, \eta_{v,v[],v[]} = 0, \tau_{t,t,t} = 0, \eta_{v,x} = \eta_{v,u[]}, \eta_{w,x} = \eta_{w,u[]}, \eta_{u,v[]} =$$

$$-\eta_{v,u[]}, \eta_{u,w[]} = -\eta_{w,u[]}, \eta_{v,w[]} = -\eta_{w,v[]}, \eta_{u,x} = \frac{\tau_t}{2} - \frac{\xi_x}{2}, \eta_{u,u[]} = \frac{\tau_t}{2}$$

$$+ \frac{\xi_x}{2}, \eta_{v,v[]} = \frac{\tau_t}{2} + \frac{\xi_x}{2}, \eta_{w,w[]} = \frac{\tau_t}{2} + \frac{\xi_x}{2} \left]$$

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E > select(has,sys5eqns1,0):pdsolve(%):sys5soln1:=simplify(%);

$$\text{sys5soln1} := \left\{ \eta_{u,t} = f_{14}(x, u[], v[], w[]) t + f_{15}(x, u[], v[], w[]) t, \eta_{v,t} = f_{11}(x, v[], w[]) t \right. \quad (78)$$

$$+ f_{12}(x, v[], w[]) u[] + f_{13}(x, v[], w[]), \eta_{w,t} = f_3(x, w[]) t + f_6(x, w[]) u[] + f_7(x, w[]) v[]$$

$$\left. + f_8(x, w[]), \tau = \frac{1}{2} c_1 t^2 + c_2 t + c_3, \xi = c_4 x + c_5 \right\}$$

**E > map(rhs,sys5soln1):indets(%,name) minus indets(symm_vars,name)
:sys5Cs:=%;indets(%%,name) minus indets(indets(%%,function),
name) minus %;**

$$\text{sys5Cs} := \{c_1, c_2, c_3, c_4, c_5\}$$

$$\{t\}$$

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E >

```

E > dsubs(sys5soln1,sys5eqns1):{seq(coeffs(expand(lhs(%[i])-rhs(%[i]))
),[t]),i=1..nops(%))}:rifsimp({%[]},casesplit):simplify(%[Solved]
):sys5eqns2:=%;nops(%);

```

$$\begin{aligned}
\text{sys5eqns2} := & \left[f_{11_x} = 0, f_{12_x} = 0, f_{13_x} = f_{12}(x, v[], w[]), f_{14_x} = \frac{c_1}{2}, f_{15_x} = \frac{c_2}{2} - \frac{c_4}{2}, f_{3_x} = 0, f_{6_x} = 0, f_{7_x} = 0, \right. \\
& f_{8_x} = f_6(x, w[]), f_{14_{u[]}} = \frac{c_1}{2}, f_{15_{u[]}} = \frac{c_2}{2} + \frac{c_4}{2}, f_{11_{v[]}} = \frac{c_1}{2}, f_{12_{v[]}} = 0, f_{13_{v[]}} = \frac{c_2}{2} + \frac{c_4}{2}, f_{14_{v[]}} \\
& = 0, f_{15_{v[]}} = -f_{12}(x, v[], w[]), f_{11_{w[]}} = 0, f_{12_{w[]}} = 0, f_{13_{w[]}} = -f_7(x, w[]), f_{14_{w[]}} = 0, f_{15_{w[]}} = \\
& \left. -f_6(x, w[]), f_{3_{w[]}} = \frac{c_1}{2}, f_{6_{w[]}} = 0, f_{7_{w[]}} = 0, f_{8_{w[]}} = \frac{c_2}{2} + \frac{c_4}{2} \right]
\end{aligned}$$

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(80)

```

E > select(has,sys5eqns2,0);pdsolve(%);subs({c__1=c__11,c__2=c__12,
c__3=c__13,_F15(v[])=_F25(v[])},%):sys5soln2:=%;

```

$$[f_{11_x} = 0, f_{12_x} = 0, f_{3_x} = 0, f_{6_x} = 0, f_{7_x} = 0, f_{12_{v[]}} = 0, f_{14_{v[]}} = 0, f_{11_{w[]}} = 0, f_{12_{w[]}} = 0, f_{14_{w[]}} = 0, f_{6_{w[]}} = 0, f_{7_{w[]}} = 0]$$

$$\{f_{11}(x, v[], w[]) = f_{15}(v[]), f_{12}(x, v[], w[]) = c_1, f_{14}(x, u[], v[], w[]) = f_{16}(x, u[]), f_3(x, w[]) = f_{17}(w[]), f_6(x, w[]) = c_2, f_7(x, w[]) = c_3\}$$

$$\text{sys5soln2} := \{f_{11}(x, v[], w[]) = _F25(v[]), f_{12}(x, v[], w[]) = c_{11}, f_{14}(x, u[], v[], w[]) = f_{16}(x, u[]), f_3(x, w[]) = f_{17}(w[]), f_6(x, w[]) = c_{12}, f_7(x, w[]) = c_{13}\} \quad (81)$$

```

E > map(rhs,sys5soln2):indets(%,name) minus indets(symm_vars,name)
:sys5Cs1:=%;indets(%%,name) minus indets(indets(%%,function),
name) minus %;

```

$$\text{sys5Cs1} := \{c_{11}, c_{12}, c_{13}\}$$

$\emptyset$

(82)

E >

```

E > dsubs(sys5soln2,sys5eqns2):rifsimp({%[]},casesplit):simplify(%
[Solved]):sys5eqns3:=%;nops(%);

```

$$\begin{aligned}
\text{sys5eqns3} := & \left[f_{13_x} = c_{11}, f_{15_x} = \frac{c_2}{2} - \frac{c_4}{2}, f_{16_x} = \frac{c_1}{2}, f_{8_x} = c_{12}, f_{15_{u[]}} = \frac{c_2}{2} + \frac{c_4}{2}, f_{16_{u[]}} = \frac{c_1}{2}, f_{13_{v[]}} \right. \\
& = \frac{c_2}{2} + \frac{c_4}{2}, f_{15_{v[]}} = -c_{11}, _F25_{v[]} = \frac{c_1}{2}, f_{13_{w[]}} = -c_{13}, f_{15_{w[]}} = -c_{12}, f_{17_{w[]}} = \frac{c_1}{2}, f_{8_{w[]}} = \frac{c_2}{2} \\
& \left. + \frac{c_4}{2} \right]
\end{aligned}$$

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(83)


```
E > pdsolve(sys5eqns3):sys5soln3:=simplify(%);
```

$$\text{sys5soln3} := \left\{ \begin{aligned} -F25(v[]) &= \frac{c_1 v[]}{2} + c_{19} f_{13}(x, v[], w[]) = \frac{(c_2 + c_4) v[]}{2} + c_{11} x - c_{13} w[] + c_{15}, \\ f_{15}(x, u[], v[], w[]) &= \frac{(c_2 + c_4) u[]}{2} + \frac{x c_2}{2} - \frac{c_4 x}{2} - c_{11} v[] - c_{12} w[] + c_{14} f_{16}(x, u[]) \\ &= \frac{(x + u[]) c_1}{2} + c_{17} f_{17}(w[]) = \frac{c_1 w[]}{2} + c_{18} f_8(x, w[]) = \frac{(c_2 + c_4) w[]}{2} + c_{12} x + c_{16} \end{aligned} \right\} \quad (84)$$

```
E > map(rhs,sys5soln3):indets(% ,name) minus indets({k_soln5,
symm_vars},name):sys5Cs2:=% union sys5Cs1;
sys5Cs2 := {c1, c11, c12, c13, c14, c15, c16, c17, c18, c19, c2, c4}
```

(85)

```
E > # all eqns solved
```

```
E >
```

```
E > sys5soln1:dsubs(sys5soln2,%):dsubs(sys5soln3,%):soln5:=simplify(
[%[]]);
```

$$\begin{aligned} \text{soln5} := & \left[\begin{aligned} \text{eta_u} &= \frac{(c_1 t + c_2 + c_4) u[]}{2} + \frac{(c_1 x + 2 c_{17}) t}{2} + \frac{(c_2 - c_4) x}{2} - c_{11} v[] - c_{12} w[] \\ &+ c_{14} \text{eta_v} = \frac{(c_1 t + c_2 + c_4) v[]}{2} + (x + u[]) c_{11} + t c_{19} - c_{13} w[] + c_{15} \text{eta_w} \\ &= \frac{(c_1 t + c_2 + c_4) w[]}{2} + (x + u[]) c_{12} + t c_{18} + c_{13} v[] + c_{16} \tau = \frac{1}{2} c_1 t^2 + c_2 t + c_3 \xi = c_4 x \\ &+ c_5 \end{aligned} \right] \end{aligned} \quad (86)$$

```
E > dsubs(soln5,detsys):dsubs(k_soln5,%);
{0}
```

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```
E > # verified
```

```
E >
```

```
E > map(rhs,soln5):indets(% ,name) minus indets({symm_vars,k_soln5},
name):soln5Cs:=%;nops(%);
soln5Cs := {c1, c11, c12, c13, c14, c15, c16, c17, c18, c19, c2, c3, c4, c5}
```

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```
E >
```

```
E > sys5_k
```

$$k(\lambda) = \frac{k0}{(\lambda + 1)^4} \quad (89)$$

```
E >
```

```
E >
```

```

E >
E > ##### Find symmetry group transformations
E > old_vars:=[u(x,t),v(x,t),w(x,t)];declare(%):
      old_vars := [u(x,t),v(x,t),w(x,t)]
      u(x,t) will now be displayed as u
      v(x,t) will now be displayed as v
      w(x,t) will now be displayed as w

```

(90)

```

E > new_vars:=[_u(_x,_t),_v(_x,_t),_w(_x,_t)];declare(%):
      new_vars := [_u(_x,_t),_v(_x,_t),_w(_x,_t)]
      _u(_x,_t) will now be displayed as _u
      _v(_x,_t) will now be displayed as _v
      _w(_x,_t) will now be displayed as _w

```

(91)

```

E > infts:=[tau(dep)=_xi[t],xi(dep)=_xi[x],eta_u(dep)=_eta[u],eta_v
      (dep)=_eta[v],eta_w(dep)=_eta[w]];
      infts := [tau = _xi, xi = _xi, eta_u = _eta, eta_v = _eta, eta_w = _eta]

```

(92)

```

E >
E > with(ListTools):
E >
E > ### CASE 1
E > dim1:=nops(soln1Cs);
      dim1 := 12

```

(93)

```

E >
E >
E > subs(infts,soln1):[seq(Reverse([seq(lhs(%[i])=coeff(rhs(%[i]),
      soln1Cs[j]),i=1..nops(infts))]),j=1..dim1)]:[seq([soln1Cs[i],
      [SymmetryTransformation(%[i],old_vars,new_vars)[]]],i=1..nops(%))
      ]:subs({RootOf(_Z^2*sec(_epsilon)^2 - 1)=cos(_epsilon),RootOf
      (_Z^2 - cos(_epsilon)^2)=cos(_epsilon),RootOf(2*_Z^2 - cos(2*
      _epsilon) - 1)=cos(_epsilon)},%):simplify(expand(%)):subs
      (_epsilon=epsilon,%):symm_generators1:=sort(%,length):simplify
      (expand(%)):subs(_epsilon=epsilon,%):symm_generators1:=sort(%,
      length):
E > seq(print(symm_generators1[i]),i=1..dim1);
      [c2,[_t=t,_x=x+epsilon,_u=u,_v=v,_w=w]]
      [c13,[_t=t,_x=x,_u=u,_v=v+epsilon,_w=w]]
      [c14,[_t=t,_x=x,_u=u+epsilon,_v=v,_w=w]]
      [c15,[_t=t,_x=x,_u=u,_v=v,_w=w+epsilon]]
      [c16,[_t=t+epsilon,_x=x,_u=u,_v=v,_w=w]]
      [c3,[_t=t,_x=x,_u=t*epsilon+u,_v=v,_w=w]]
      [c4,[_t=t,_x=x,_u=u,_v=v,_w=t*epsilon+w]]
      [c11,[_t=t,_x=x,_u=u,_v=t*epsilon+v,_w=w]]

```

$$\begin{aligned}
& [c_7, [_t = e^\epsilon t, _x = x e^\epsilon, _u = e^\epsilon u, _v = e^\epsilon v, _w = e^\epsilon w]] \\
& [c_6, [_t = t, _x = x, _u = u, _v = v \cos(\epsilon) - \sin(\epsilon) w, _w = \sin(\epsilon) v + w \cos(\epsilon)]] \\
& [c_5, [_t = t, _x = x, _u = \cos(\epsilon) u - \sin(\epsilon) w + (\cos(\epsilon) - 1) x, _v = v, _w = u \sin(\epsilon) + w \cos(\epsilon) \\
& \quad + \sin(\epsilon) x]] \\
& [c_{12}, [_t = t, _x = x, _u = \cos(\epsilon) u - \sin(\epsilon) v + (\cos(\epsilon) - 1) x, _v = u \sin(\epsilon) + v \cos(\epsilon) + \sin(\epsilon) x, _w \\
& \quad = w]] \tag{94}
\end{aligned}$$

E >

E >

E > ### CASE 3

E > dim3:=nops(soln3Cs);

dim3 := 13

(95)

E >

E >

E > subs(infts,soln3):[seq(Reverse([seq(lhs(%[i])=coeff(rhs(%[i]),
soln3Cs[j]),i=1..nops(infts))]),j=1..dim3)]:[seq([soln3Cs[i],
[SymmetryTransformation(%[i],old_vars,new_vars)[]]],i=1..nops(%))
]:subs({RootOf(_Z^2*sec(_epsilon)^2 - 1)=cos(_epsilon),RootOf(
_Z^2 - cos(_epsilon)^2)=cos(_epsilon),RootOf(2*_Z^2 - cos(2*
_epsilon) - 1)=cos(_epsilon)},%):simplify(expand(%)):subs
(_epsilon=epsilon,%):symm_generators3:=sort(%,length):simplify
(expand(%)):subs(_epsilon=epsilon,%):symm_generators3:=sort(%,
length):

E > seq(print(symm_generators3[i]),i=1..dim3);

$$\begin{aligned}
& [c_2, [_t = t + \epsilon, _x = x, _u = u, _v = v, _w = w]] \\
& [c_4, [_t = t, _x = x + \epsilon, _u = u, _v = v, _w = w]] \\
& [c_{15}, [_t = t, _x = x, _u = u, _v = v + \epsilon, _w = w]] \\
& [c_{16}, [_t = t, _x = x, _u = u, _v = v, _w = w + \epsilon]] \\
& [c_{24}, [_t = t, _x = x, _u = u + \epsilon, _v = v, _w = w]] \\
& [c_{14}, [_t = t, _x = x, _u = u, _v = v, _w = t\epsilon + w]] \\
& [c_{11}, [_t = t, _x = x, _u = t\epsilon + u, _v = v, _w = w]] \\
& [c_{12}, [_t = t, _x = x, _u = u, _v = t\epsilon + v, _w = w]] \\
& [c_6, [_t = t, _x = x, _u = u, _v = v \cos(\epsilon) - \sin(\epsilon) w, _w = \sin(\epsilon) v + w \cos(\epsilon)]] \\
& \left[c_7, \left[_t = e^\epsilon t, _x = x, _u = (x + u) e^{-\frac{2\epsilon}{P}} - x, _v = e^{-\frac{2\epsilon}{P}} v, _w = e^{-\frac{2\epsilon}{P}} w \right] \right] \\
& [c_5, [_t = t, _x = x, _u = \cos(\epsilon) u - \sin(\epsilon) w + (\cos(\epsilon) - 1) x, _v = v, _w = u \sin(\epsilon) + w \cos(\epsilon) \\
& \quad + \sin(\epsilon) x]] \\
& [c_{13}, [_t = t, _x = x, _u = \cos(\epsilon) u - \sin(\epsilon) v + (\cos(\epsilon) - 1) x, _v = u \sin(\epsilon) + v \cos(\epsilon) + \sin(\epsilon) x, _w \\
& \quad = w]]
\end{aligned}$$

$$\left[c_3, \left[_t=t, _x=x e^\epsilon, _u=(x+u) e^{\frac{\epsilon(p+2)}{p}} - x e^\epsilon, _v=v e^{\frac{\epsilon(p+2)}{p}}, _w=w e^{\frac{\epsilon(p+2)}{p}} \right] \right] \quad (96)$$

E >

E >

E >

E > ### CASE 5

E > dim5:=nops(soln5Cs);

dim5 := 14

(97)

E >

E >

E > subs(infts,soln5):[seq(Reverse([seq(lhs(%[i])=coeff(rhs(%[i]),
soln5Cs[j]),i=1..nops(infts))]),j=1..dim5)]:[seq([soln5Cs[i],
[SymmetryTransformation(%[i],old_vars,new_vars)[[]]],i=1..nops(%))
]:subs({RootOf(_Z^2*sec(_epsilon)^2 - 1)=cos(_epsilon),RootOf(
_Z^2 - cos(_epsilon)^2)=cos(_epsilon),RootOf(2*_Z^2 - cos(2*
_epsilon) - 1)=cos(_epsilon)},%):simplify(expand(%),symbolic)
:subs(_epsilon=epsilon,%):symm_generators5:=sort(%,length)
:simplify(expand(%)):subs(_epsilon=epsilon,%):symm_generators5:=
sort(%,length):

E > seq(print(symm_generators5[i]),i=1..dim5);

$[c_3, [_t=t+\epsilon, _x=x, _u=u, _v=v, _w=w]]$

$[c_5, [_t=t, _x=x+\epsilon, _u=u, _v=v, _w=w]]$

$[c_{14}, [_t=t, _x=x, _u=u+\epsilon, _v=v, _w=w]]$

$[c_{15}, [_t=t, _x=x, _u=u, _v=v+\epsilon, _w=w]]$

$[c_{16}, [_t=t, _x=x, _u=u, _v=v, _w=w+\epsilon]]$

$[c_{17}, [_t=t, _x=x, _u=t\epsilon+u, _v=v, _w=w]]$

$[c_{18}, [_t=t, _x=x, _u=u, _v=v, _w=t\epsilon+w]]$

$[c_{19}, [_t=t, _x=x, _u=u, _v=t\epsilon+v, _w=w]]$

$[c_{13}, [_t=t, _x=x, _u=u, _v=v\cos(\epsilon) - \sin(\epsilon)w, _w=\sin(\epsilon)v + w\cos(\epsilon)]]$

$\left[c_2, \left[_t=e^\epsilon t, _x=x, _u=(x+u) e^{\frac{\epsilon}{2}} - x, _v=v e^{\frac{\epsilon}{2}}, _w=w e^{\frac{\epsilon}{2}} \right] \right]$

$\left[c_\phi, \left[_t=t, _x=x e^\epsilon, _u=(x+u) e^{\frac{\epsilon}{2}} - x e^\epsilon, _v=v e^{\frac{\epsilon}{2}}, _w=w e^{\frac{\epsilon}{2}} \right] \right]$

$\left[c_1, \left[_t=-\frac{2t}{t\epsilon-2}, _x=x, _u=\frac{-tx\epsilon-2u}{t\epsilon-2}, _v=-\frac{2v}{t\epsilon-2}, _w=-\frac{2w}{t\epsilon-2} \right] \right]$

$[c_{11}, [_t=t, _x=x, _u=\cos(\epsilon)u - \sin(\epsilon)v + (\cos(\epsilon) - 1)x, _v=u\sin(\epsilon) + v\cos(\epsilon) + \sin(\epsilon)x, _w=w]]$

$[c_{12}, [_t=t, _x=x, _u=\cos(\epsilon)u - \sin(\epsilon)w + (\cos(\epsilon) - 1)x, _v=v, _w=u\sin(\epsilon) + w\cos(\epsilon) + \sin(\epsilon)x]]$ (98)

E >